range of the histogram. This is because it requires all three colour channels to reach their maximum for clipping, as opposed to just one of the three channels for an RGB histogram.

5. Desaturating the colours in an image is the simplest type of conversion, but often produces inadequate results. This is because it does not allow for control over how the primary colours combine to produce a given greyscale brightness. Despite this, it is probably the most commonly used way of converting into black and white. In Photoshop, this is accomplished by going from `Image > Adjustments > Desaturate`. Ordinarily, the best results are achieved when the image has the correct white balance. Removal of colour casts means that the colours will be more pure, and so the results of any colour filter will be more pronounced.

Any black and white conversion which utilizes a significant boost in colour saturation may begin to show artefacts, such as increased noise, clipping or loss of texture detail. On the other hand, higher colour saturations also mean that each colour filter will have a more pronounced effect.

Recall that the noise levels in each colour channel can be quite different, with the blue and green channels having the most and lease noise, respectively. Try to use as little of the blue channel as possible to avoid excess noise.

Levels and curves can be used in conjunction with black and white conversion to provide further control over tones and contrast. Keep in mind though, that some contrast adjustments can only be made by choosing an appropriate colour filter, since this adjusts relative contrast within and the colour regions. Care should also be taken when using these because even slight colour clipping in any of the individual colour channels can become quite apparent in black and white depending on which channels are used for conversion.